

Date: 6th March 2017 Revision: 1.3 Resp: ARM Teamlead Page: 1 of 9	Detailed Design Document	
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ZF - Division A

Slip Control Software Design

Document ID	:	ARM-DDD-Arm_Ay_Mod
Title	:	ARM-DDD-Arm_Ay_Mod
Summary	:	Description of subsystem architecture and interface
Scope	:	Application: • arm_ay_mod{.c, _priv.h, _sub.h} Calibration:
Doc Template	:	DDD-Module-Template based on Template C43-03-006-SCS-EN-F01 Issue B
Project Name	:	AR 16 Core Development
Doc Status	:	Draft / For Review / Final / Released
Audience	:	Only for TRW internal use.

Release Date	Author	Reviewer	Approved by
	A. Hoehn		

REFERENCES

Ref.	Document	Title	Ver.	Location

REVISION HISTORY

Rev.	Date dd/mm/yyyy	Author	Change Request	Comments (Version)
1.0	30/05/2011	B. Etges		Initial Version
1.2	23/08/2016	A. Hoehn		Update to AR 16
1.3	06/03/2017	A. Hoehn	1800840	WTSA A _y Sync, ZF Logo

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1 ABSTRACT

This module calculates the base signal [AyMod](#) of the [ARM](#) controller. The signal [AyMod](#) predicts the lateral acceleration in the soon future. [AyMod](#) is of special interest since the [ARM](#) pressure request is directly linked to [AyMod](#).

2 ACRONYMS

ARM Active Roll Mitigation

AyMod 'Modified' Lateral Acceleration, that tries to predict the Lateral Acceleration

Swa Steering Wheel Angle

Rwa Road Wheel Angle

WTSA Wheel Transition Steering Amplification

WTP Wheel Transition Pressure

3 OVERVIEW

This document describes the purpose and behavior of Arm_Ay_Mod.

3.1 DESIGN GOAL

The predicted lateral acceleration is calculated based on the following data.

- The actual lateral acceleration.
- The steering input by the driver.
- The estimated surface friction.

3.2 GENERAL REQUIREMENTS

1. [ARM](#) should prevent dynamic rollover.
2. [ARM](#) should prevent static rollover.

3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

This module needs only to be processed, when [ARM](#) is allowed.

3.5 FLOW-CHARTS AND DIAGRAMS

The calculation of [AyMod](#) is presented in Fig. 3.1 below.

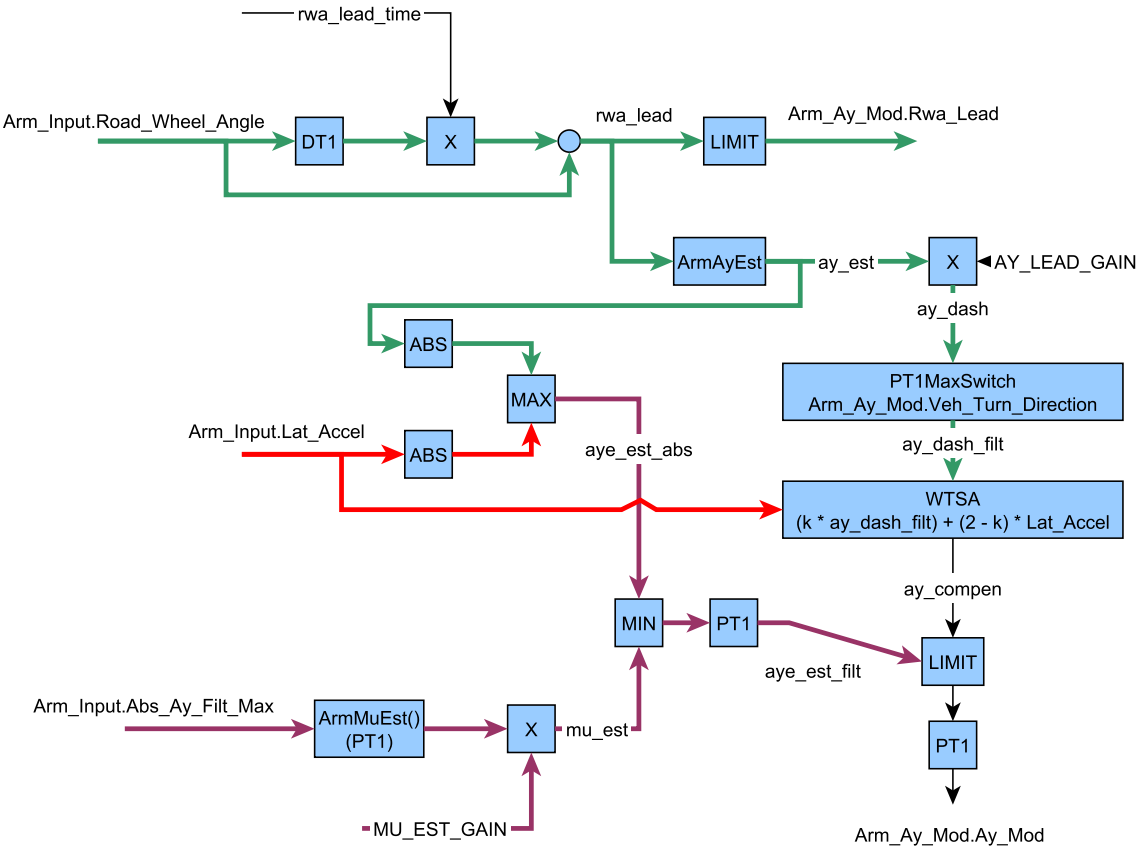


Figure 3.1: Calculation of [AyMod](#)

As mentioned in Section 3.1, [AyMod](#) is influenced by the driver steering input (green), the measured lateral acceleration (red) and the estimated friction (purple).

4 DETAILED DESIGN

This chapter outlines the detailed design of Arm_Ay_Mod.

4.1 ALGORITHM DESCRIPTION

AyMod is calculated from a predicted lateral acceleration, that is limited depending on surface friction. Since both steering wheel angle and road wheel angle represent the steering input, these names will be used synonymously in the following sections.

4.1.1 Prediction of Lateral Acceleration

The lateral acceleration is predicted by using a single track vehicle model, that uses a predicted steering wheel input.

Predicted Steering Angle

To predict the steering input in the future, the derivative is used as a preview. It is directly added to the original signal which yields:

$$\text{swa_lead} = \text{swa} + \text{TRIM} \cdot \frac{d\text{swa}}{dt} \quad (4.1)$$

Lateral Acceleration based on Predicted Steering Input

A single track model is used to predict the lateral acceleration caused by the steering input.

$$\begin{aligned} \text{ay_est} &= f(\text{swa_lead}) \\ &= \frac{v^2 \cdot \text{swa_lead}}{i \cdot l \cdot (1 + (\frac{v}{v_{ch}})^2)} \end{aligned} \quad (4.2)$$

$$\text{ay_compen} = \text{ay_dash_filt} + \text{Lat_Accel} \quad (4.3)$$

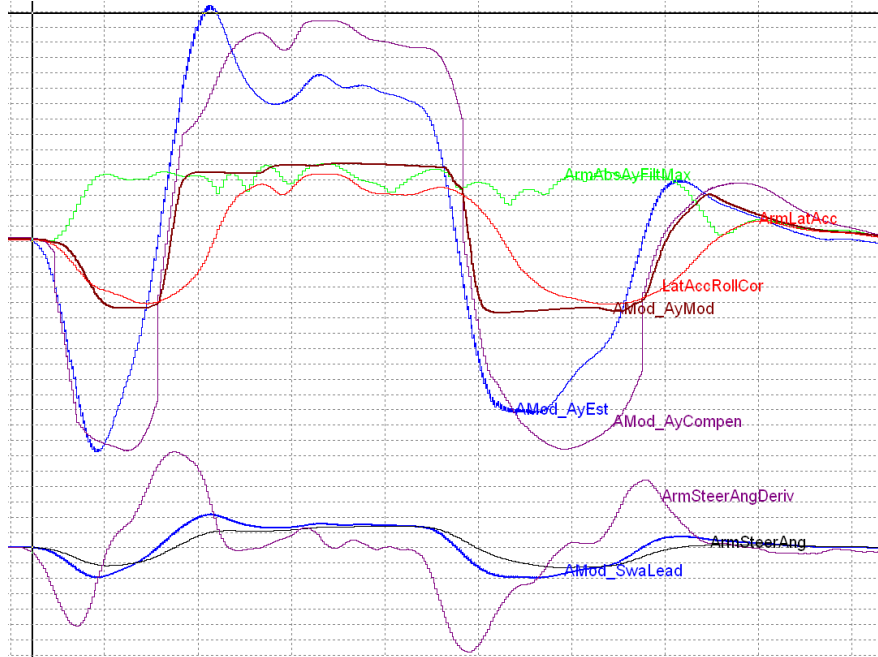


Figure 4.1: **AyMod** signals without WTSA

Wheel Transition Steering Amplification

WTSA is an optional feature during **AyMod** calculation. It is helpful in lane change maneuvers. This logic starts, when `Arm_Ay_Mod.Veh_Turn_Direction` changes sign and the driver steers to the opposite lane. The amplification should be faded out to neutral $WTSA\text{-factor} = 100\%$, as soon as the lateral acceleration changes sign.

There are two main goals:

1. During wheel transition phase, `Arm_Input.Lat_Accel` is still on the old side, but A_y should not increase anymore at the old side.
2. If **WTSA** is active, **AyMod** is governed by the steering input.

During the **AyMod** calculation, **WTSA** affects `ay_compen`. Hence

$$ay_compen = (WTSA\text{-factor} \cdot ay_dash_filt) + (200\% - WTSA\text{-factor}) \cdot Lat_Accel \quad (4.4)$$

To prevent braking the wrong wheel, the measured A_y signal should not differ too much from the latched A_y signal.

Fig. 4.2 presents the **WTSA** influence on **Ay_Mod**.

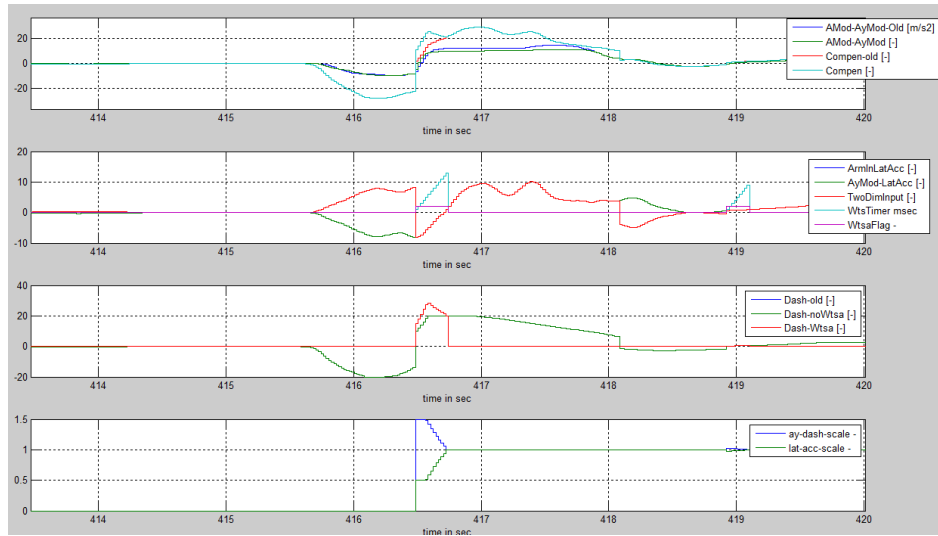


Figure 4.2: **AyMod** signals with **WTSA**

Remark: **WTSA** affects **WTP** since **WTSA** forces `Arm_State.Turn_Direction` to change sign earlier. Therefore the time between `Arm_Ay_Mod.Veh_Turn_Direction` and `Arm_State.Turn_Direction` is small and there is almost no time for prefill on opposite wheel (**WTP**).

4.1.2 Estimation of surface friction limit

The surface friction estimation uses a predicted lateral acceleration (`Ay_Abs_HiFilt`) that is calculated in **ARM**. If the lateral acceleration increases and the driver steers in the same

direction, the lateral acceleration is increased with a derivative part depending on driver steering velocity.

$$A_{y_Abs_HiFilt} = ABS\left(\frac{d}{dt} \cdot Lat_Accel\right) \cdot f(ABS(A_{y_From_Swa_Lead})) + ABS(Lat_Accel) \quad (4.5)$$

This signal is rate limited during **ARM** intervention and filtered in order to enable fast rise and very slowly decay rates.

Depending on the configuration `ENABLE_AYMOD_MU_EST_MIN_AY == FALSE`, this signal is used to limit **AyMod** in the next step (`aye_est_filt`). This configuration is often used in very heavy vehicles with high roll over propensity, because **AyMod** stays nearly constant at a high level until the end.

In lighter vehicles with less roll over propensity, it is possible to set the configuration `ENABLE_AYMOD_MU_EST_MIN_AY` to `TRUE`, so that a decreasing measured lateral acceleration would also decrease the **AyMod** signal.

$$aye_est_filt = MIN(mu_est, MAX(ABS(ay_est), ABS(Lat_Accel))) \quad (4.6)$$

4.1.3 Calculation of **AyMod**

AyMod is the friction limited `ay_compen`. See Fig. 4.3 for an example of the friction limitation. The influence can be observed during side change on `AMod_Aye_Est_Filt` (solid green line).

$$AyMod = LIMIT(ay_compen, -aye_est_filt, aye_est_filt) \quad (4.7)$$

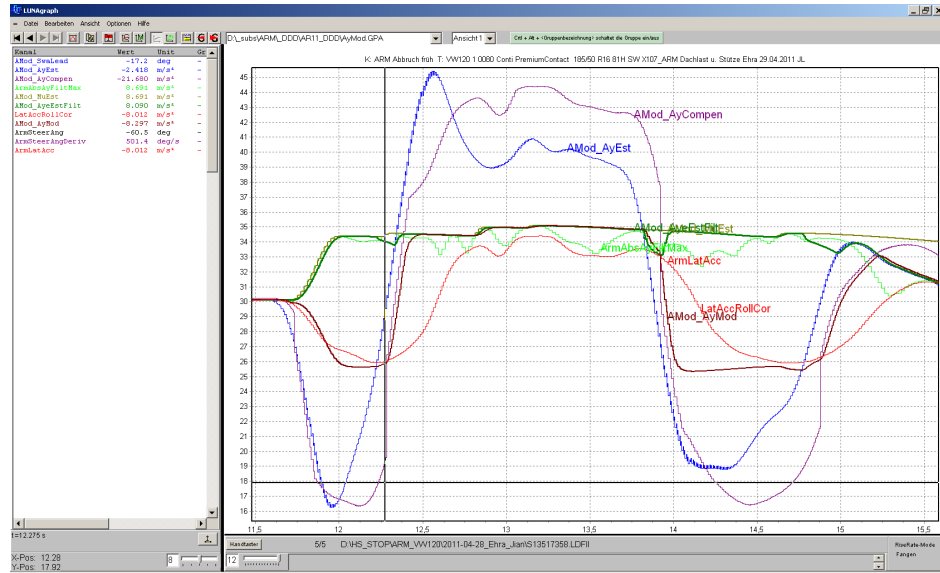


Figure 4.3: **AyMod** and `ENABLE_AYMOD_MU_EST_MIN_AY` set to `TRUE`

5 PROVIDED TASKS

5.1 ArmAyMod

Description	ArmAyMod
Rate	Standard Arm timebase 5 - 20ms
Dependency	
SYNTAX	void ArmAyMod (void)

6 PROVIDED SERVICES

- ArmAyEst

6.1 Private Services

- ArmMuEst
- WTSteerAmplification

7 CONFIGURATION OPTIONS

Name	Type	Range	Description
ENABLE_AYMOD_MU_EST_MIN_AY	BOOLEAN	[0...1]	En-/Disable decreasing AyMod due to decreased Lat_Accel

8 CONFIGURATION PROCEDURES

Depending on project requirements and the roll over propensity of the vehicle the configurations has to be adjusted once before start of project. Changes during development phase of the project can be applied but should be carefully reviewed with the subsystem responsible Teamlead.

9 TEST CASES

Test category	
Requirements	
Test Case ID	

10 CALIBRATION INFORMATION

AyMod is an important signal in **ARM**, since both activation and pressure request strongly rely on it.

10.1 Signal Processing

It is advised to keep the core trimming for the signal processing part.

10.2 Wheel Transition Steering Amplification (**WTSA**)

WTSA can be deactivated by setting **AYMOD_WTSA_MAX_TIME** to zero, since this parameter defines how long the transition phase is supposed to last. During transition phase the behavior is determined by **AYMOD_WTSA_TBL**. If **AyMod** changes sign to slowly during a lane change maneuver, try to decrease **AYMOD_WTSA_TBL_X0** to start the amplification earlier or increase **AYMOD_WTSA_TBL_Y0** to obtain a stronger amplification.

As stated in the Detailed Design, the lateral acceleration is latched during wheel transition if **WTSA** is active. To prevent braking the wrong wheel, the measured A_y signal should not differ too much from the latched A_y signal. The allowed difference can be adjusted by **AYMOD_WTSA_HOLD_LIMIT**. If the difference exceeds this limit, the latched signal is synchronized to the measured one, see Fig. 10.1.

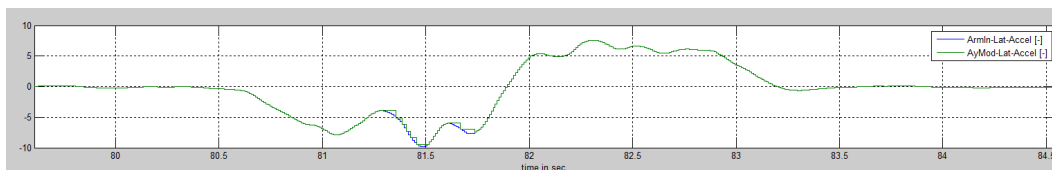


Figure 10.1: Lat_Accel Sync during **WTSA**

11 APPENDIX